

KANBAN TASK MANAGEMENT WEBSITE

Mini project Report

Submitted in partial fulfillment of the requirement of
University of Mumbai for the Degree of
(Computer Engineering)

By

Subodh Kumar Sahu

ID NO: TU3F2223154

Shruti Parange

ID No: TU3F2223143

Sarika Katkar

ID No: TU3F2223117

Moinuddin Quazi

ID No: TU3F2223193

Under the Guidance of

Prof. Preeti Patil



Department of Computer Engineering

TERNA ENGINEERING COLLEGE

Plot no.12, Sector-22, Opp. Nerul Railway station,

Phase-11, Nerul (w), Navi Mumbai 400706

UNIVERSITY OF MUMBAI

(2024-2025)



TERNA ENGINEERING COLLEGE, NERUL, NAVI MUMBAI
Department of Computer Engineering

Academic Year 2024-25

CERTIFICATE

This is to certify that the mini project entitled “**KANBAN TASK MANAGEMENT WEBSITE**”
is a bonafide work of

Subodh Kumar Sahu

ID NO: TU3F2223154

Shruti Parange

ID No: TU3F2223143

Sarika Kakar

ID No: TU3F2223117

Moinuddin Quazi

ID No: TU3F2223193

Submitted to the University of Mumbai in partial fulfillment of the requirement for the
award of the Bachelor of Engineering (Computer Engineering).

Guide

Head of Department

Principal

Project Report Approval

This Mini Project 2B Report – entitled “**KANBAN TASK MANAGEMENT WEBSITE**” by following students is approved for the degree of *B.E. in "Computer Engineering"*.

Submitted by:

Subodh Kumar Sahu

ID NO: TU3F2223154

Shruti Parange

ID No: TU3F2223143

Sarika Katkar

ID No: TU3F2223117

Moinuddin Quazi

ID No: TU3F2223193

Examiners Name & Signature:

1. _____

2. _____

Date:

Place:

Declaration

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Subodh Kumar Sahu	ID NO: TU3F2223154	-----
Shruti Parange	ID NO: TU3F2223143	-----
Sarika Katkar	ID NO: TU3F2223117	-----
Moinuddin Quazi	ID No: TU3F2223193	-----

Date:

Place:

Acknowledgement

We would like to express our sincere gratitude towards our guide **Prof. Preeti Patil**, for her help, guidance and encouragement, she provided during the project development. This work would have not been possible without her valuable time, patience and motivation. We thank her for making our stint thoroughly pleasant and enriching. It was great learning and an honor being their student.

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We take the privilege to express our sincere thanks to **Dr. L. K. Ragha is our principal** for providing encouragement and much support throughout our work.

Subodh Kumar Sahu	ID NO: TU3F2223154	_____
Shruti Parange	ID NO: TU3F2223143	_____
Sarika Katkar	ID NO: TU3F2223117	_____
Moinuddin Quazi	ID No: TU3F2223193	_____

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Index

TABLE OF CONTENTS

Chapter	Title	Pg No.
	Abstract	1
	List of Figures	2
	List of Tables	2
1	Introduction	3
	1.1 Motivation	4
	1.2 Need of problem	4
	1.3 Scope of Project	5
	1.4 Aim	7
	1.5 Objective	8
2	Literature Review	10-12
3	3.1 Problem Statement	13
	3.2 Features	13
	3.3 Specifications of the System	14
4	Design and Implementation	17
	Hosting on AWS-S3	19
5	5.1 Conclusion	26
	5.2 Future Scope	26
	References	28

Abstract

This project involves the development and deployment of a Kanban-based Task Management Website designed to streamline task tracking and team collaboration using a visual board layout. The application enables users to create, assign, move, and monitor tasks across customizable columns (e.g., To-Do, In Progress, Done), enhancing workflow visibility and productivity.

The system is built using modern web technologies (such as Node.js, React.js, and MongoDB) and is securely deployed on AWS cloud infrastructure. Key AWS services used include EC2 (for hosting), S3 (for static file storage), and IAM (for access control). The application is containerized using Docker and orchestrated for scalability and reliability.

By leveraging AWS, the platform ensures high availability, scalability, and performance, making it ideal for both small teams and large enterprise environments. This project demonstrates practical implementation of cloud deployment, DevOps, and task automation principles in a real-world application.

List of Figures

Sr No.	Title	Pg No.
1	Website interface	17
2	Adding new task option	17
3	Adding new board option	18
4	Delete Board option	18
5	Create S3 Bucket	19
6	Bucket Successfully created	20
7	Upload project files to bucket	20
8	Project files uploaded successfully	21
9	Making changes in static website hosting page	21
10	Enable Static website hosting	22
11	Edit Bucket policy and generate policy for website	22
12	Generating policy for website	23
13	Policy generated successfully	24
14	Add policy generated and save changes	24
15	Policy added successfully	25
16	Final URL of website is created	25
17	Website is hosted on AWS server with the help of S3 bucket	26

List of Tables

Sr No.	Title	Pg No.
1	Literature Survey	10-12

Chapter 1

Introduction

The Kanban Task Management Website is a cloud-based project management solution designed to help individuals and teams organize, track, and manage their tasks visually using the Kanban methodology. The platform enables users to create boards, lists (columns), and tasks (cards), allowing for a flexible and intuitive workflow that reflects real-time project progress.

The application is developed using a modern tech stack (React.js, Node.js, MongoDB) and is hosted on AWS Cloud Infrastructure, ensuring high availability, scalability, and security. With features like drag-and-drop task movement, user roles, real-time updates, and secure authentication, the system enhances productivity and collaboration across distributed teams.

The backend services are containerized using Docker and deployed on Amazon EC2 instances, while static assets are served through Amazon S3. Additional services like AWS IAM, Route 53, and CloudWatch are used to manage access, route traffic, and monitor performance.

1.1 Motivation

In today's fast-paced and collaborative work environments, managing tasks effectively is crucial to the success of teams and organizations. Traditional task management tools often lack flexibility, real-time collaboration, and scalability, making it difficult for teams to track and complete projects efficiently—especially when working remotely or across time zones.

The motivation for developing a **Kanban Task Management Website** stems from the need for a lightweight, user-friendly, and cloud-accessible solution that allows individuals and teams to visualize their workflow, prioritize tasks, and collaborate seamlessly. By adopting the Kanban methodology, users can easily manage tasks across customizable stages such as *To-Do*, *In Progress*, *Review*, and *Done*.

Hosting the application on **AWS Cloud Infrastructure** further enhances the platform by providing:

- **Global accessibility**
- **Scalability on demand**
- **Improved performance**
- **High availability**
- **Strong security controls**

This project combines the flexibility of Kanban boards with the power of cloud computing, offering a robust, real-time task management platform.

1.2 Need of the Problem

Many teams rely on outdated or non-scalable tools that do not support modern project management needs. The main challenges faced by users in traditional systems include:

- **Lack of real-time updates:** Teams are unable to see live task progress or recent changes.
- **Poor collaboration:** Limited options for sharing tasks, comments, or assigning team roles.
- **No customization:** Fixed workflows that don't adapt to different team processes.
- **High maintenance:** On-premise systems require setup, maintenance, and infrastructure.
- **Low scalability:** Systems may not handle large teams or multiple projects effectively

The Kanban Task Management Website addresses these issues by offering a cloud-native and collaborative environment where users can:

- Create and manage projects from anywhere
- Customize workflows according to their process
- Assign, update, and track tasks in real time
- Use secure and scalable cloud services like AWS EC2, S3, and IAM
- Benefit from auto-scaling, load balancing, and cloud-based monitoring

This solves a major gap in scalable, secure, and efficient project tracking tools, especially for remote-first and agile teams.

1.3 Scope of the Project

The Kanban Task Management Website aims to provide a fully functional and scalable task management system deployed on Amazon Web Services (AWS). The scope of the project includes both technical implementation and usability goals.

User Management & Authentication

- Secure user registration and login system
- Role-based access control (Admin, Editor, Viewer)
- Integration with authentication services (JWT, OAuth, or AWS Cognito)

Kanban Board & Task Management

- Creation of multiple boards per user
- Dynamic columns (To-Do, In Progress, Done, etc.)
- Task creation, editing, assigning, commenting, and labeling
- Drag-and-drop functionality for task movement

Cloud Deployment and Infrastructure

- Hosting frontend and backend using **AWS EC2 instances**

- Storing static content using **Amazon S3**
- Routing domain via **Amazon Route 53**
- Monitoring and performance tracking using **CloudWatch**
- Secure access management using **IAM roles and policies**

Database and Storage

- Use of **MongoDB Atlas** (or AWS DocumentDB) for storing user data, tasks, and board details
- Cloud backup and restoration support

Scalability & Optimization

- Dockerized backend and frontend services for ease of deployment
- Setup of **Auto Scaling Groups** and **Load Balancer** for high availability
- Optimized code and caching strategies to improve response time

User Experience & Interface

- Responsive and mobile-friendly UI using **React.js**
- Minimalistic and modern dashboard for ease of navigation
- Notifications, reminders, and activity logs for improved engagement

Future Scope (Optional Enhancements)

- Integration with third-party tools like Google Calendar, Slack, Trello
- Real-time collaboration using WebSockets
- Mobile app version using React Native

1.4 Aim

The **Kanban Task Management Website** aims to provide a fully functional and scalable task management system deployed on **Amazon Web Services (AWS)**. The scope of the project includes both technical implementation and usability goals.

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1.5 Objectives

I. Design a Visual Task Management Interface:

Build a Kanban-style user interface that allows users to manage tasks through drag-and-drop functionality across different stages (To-Do, In Progress, Done).

II. Enable Multi-User Collaboration:

Allow users to create shared boards, assign tasks, and collaborate with team members using comments, tags, and role-based access.

III. Implement Secure Authentication and Authorization:

Develop secure login and signup features using JWT or OAuth, and apply role-based permissions to protect data and actions.

IV. Develop and Integrate RESTful APIs:

Create a robust backend using Node.js and Express.js to handle all application logic and communication with the database.

V. Deploy Application on AWS Cloud Infrastructure:

Use AWS services such as EC2 for hosting, S3 for static file storage, and Route 53 for DNS management to ensure global accessibility and high availability.

VI. Utilize Cloud Monitoring and Logging Tools:

Integrate **Amazon CloudWatch** for real-time monitoring, logging, and performance analysis to detect errors and maintain application health.

VII. **Ensure Scalability and Fault Tolerance:**

Use **Auto Scaling** and **Elastic Load Balancers** to support high traffic and maintain service continuity.

VIII. **Use Docker for Containerization:**

Package the application using Docker for consistent deployment across environments and ease of scaling.

IX. **Create a Responsive and User-Friendly UI:**

Design a clean, intuitive interface with **React.js**, ensuring a seamless experience across desktops, tablets, and mobile devices.

X. **Promote Agile Project Management:**

Provide flexible workflows suitable for Agile or Scrum methodologies, allowing teams to improve productivity and project transparency.

Chapter 2

Literature Review

Sr No	Author	Summary
1	Grishma Hedao, Priyanka Thoke, Raksha Tabhane, Shubham Meshram, Swapnil Kumbhalkar, Prof. Mukesh Barapatre	1. The Online Task Management System (OTMS) is designed for college departments, providing an online platform for efficient day-to-day task management and communication. 2. Key users include administrators, teachers, assistants, and students, each with specific privileges for task assignment, communication, and complaint filing. 3. Implemented with technologies like Android, PHP, MySQL, and featuring modules for login, registration, messaging, notifications, task allocation, and complaint handling, the system aims to enhance overall efficiency and communication within the college
2	Victoria Bellotti, Brinda Dalal, Nathaniel Good, Peter Flynn, Daniel G. Bobrow	1. The paper explores task management practices to inform the design of a task list manager, challenging the notion that people are poor at prioritizing tasks. 2. Individuals showcase specific strategies in task management, challenging misconceptions and highlighting factors influencing completion and representations' role. 3. The research explores task management practices to inform the design of a personal to-do list manager, studying media, organizational methods, and completion strategies.

3	Marianna Spezie and Damiano Bragantini	1. Paper advocates Task Management System adoption for enhanced productivity through improved time organization, priority-setting, and work control.
		2. Crucial aspects within a production process, identified as Crucial Aspects (CAs), include prioritization, work organization, and time/resource management. 3. A systematic literature review summarized articles to offer a balanced view of the advantages, opportunities, and limitations of adopting a TMS in organizations, targeting enhanced performance and efficiency.
4	Suyash Kejriwal, Vaibhav Vishal, Aastha Gulati, Gaurav Gambhir	1. Aiming to resolve task efficiency issues, the paper proposes a user-friendly feature for comprehensive task analysis. 2. The proposed solution involves creating an app for easy task management, deadlines, and reminders, aiding users in organizing and prioritizing tasks effectively. 3. Features: address partial task completion, track efficiency, introduce Target Completion Indicator, serve as virtual reminder for goal achievement.

5	H.C.W. Lau, S.K. Tso, J.K.L. Ho	1. The paper emphasizes the need for effective company-wide information processing in manufacturing for global competitiveness 2. It introduces an Intelligent Task Management System (ITMS) using rule-based inference mechanisms and Object-Oriented Virtual Agents (OOVA) to enhance problem-solving and service support in a manufacturing information network 3. The ITMS, tested in an emulated environment, aims to improve productivity in existing agent-based systems and influence the design of future intelligent agent-based systems for manufacturing information networks.
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Chapter 3

3.1 Problem Statement

In today's dynamic and collaborative work environments, teams often struggle with disorganized task management, lack of real-time visibility, and limited collaboration tools. Existing project management solutions may be either overly complex, expensive, or lack scalability for small teams and startups. Additionally, self-hosted tools demand technical expertise and infrastructure maintenance, which increases operational overhead.

There is a need for a lightweight, cloud-native Kanban task management system that provides users with a simple, intuitive, and scalable interface for managing projects and workflows. The system should allow teams to visually track progress, assign tasks, and collaborate effectively—while ensuring secure access, real-time updates, and global availability.

This project aims to address these challenges by building a Kanban Task Management Website and deploying it on AWS Cloud Infrastructure, leveraging services like EC2, S3, IAM, and CloudWatch to deliver a secure, scalable, and cost-efficient task management solution suitable for distributed and agile teams.

3.2 Features

Core Functionality

- Customizable board views with multiple columns (To Do, In Progress, Done, etc.)
- Drag-and-drop card movement between columns
- Task cards with titles, descriptions, due dates, and priority levels
- User assignment for tasks with avatar display
- Filtering and sorting options for tasks
- Search functionality across all boards and tasks

User Experience

- Responsive design for mobile, tablet, and desktop
- Dark/light mode toggle
- Collapsible sidebar for navigation
- Board templates for quick setup
- Activity timeline showing recent changes
- Task commenting system

Organization Features

- Multiple board support for different projects
- Swimlanes to group related tasks horizontally
- Tags/labels with color coding
- Work-in-progress limits for columns
- Subtasks and checklists

3.2 Specification of the system

Hardware Requirements

The Kanban Task Management Website requires reliable cloud infrastructure (preferably AWS EC2 instances) with sufficient CPU, memory, and storage to handle multiple users and task operations in real time. For local deployment or development purposes, a system with the following minimum specs is recommended:

- **Processor:** Intel i5 or higher / AMD equivalent
- **RAM:** 8 GB minimum
- **Storage:** 256 GB SSD
- **Network:** Stable internet connection (10 Mbps or higher)

Software Requirements

The software stack includes the following:

- **Frontend Framework:** React.js with Redux
- **Cloud Platform:** Amazon Web Services (EC2, S3, Route 53, IAM, CloudWatch)
- **Version Control:** Git + GitHub

Backup and Recovery

- **Data Backup:** Daily backups of task data, user information, and project configurations stored securely in Amazon S3.
- **Disaster Recovery:** Recovery plan using snapshots and EC2 recovery scripts to restore service within a short downtime window.

Other Non-functional Requirements

Security Requirements

- **Encryption:** TLS encryption for data transmission; optional AES encryption for data at rest.
- **Authentication:** Secure login with JWT tokens; support for OAuth2 and optional multi-factor authentication (MFA).
- **Authorization:** Role-based access control (Admin, Member, Viewer).
- **Vulnerability Management:** Regular dependency updates and penetration testing.
- **Intrusion Detection:** Use AWS CloudWatch and GuardDuty to monitor suspicious behavior.
- **Backup Security:** Encrypted backups stored in Amazon S3 with restricted IAM access.
- **Incident Response:** Defined protocol to handle security incidents including rollback and notification.

Portability Requirements

- Cross-Platform Compatibility: Accessible via modern browsers (Chrome, Firefox, Safari, Edge) on Windows, macOS, Linux, and mobile devices.
- Containerized Deployment: Easily deployable on any environment using Docker and Docker Compose.
- Standardized Components: Use of standardized RESTful APIs and cloud-compatible services for portability.

Reliability Requirements

- Redundancy: Load balancers and multi-zone deployment on AWS.
- Error Handling: Robust error handling and failover logic.
- Monitoring & Alerts: AWS CloudWatch for logs, alarms, and performance metrics.
- Data Integrity: Consistent state management using Redux and database integrity checks.

Usability Requirements

- Intuitive Interface: Clean, drag-and-drop Kanban UI with tooltips and on-screen instructions.
- Accessibility: Supports keyboard navigation, screen readers, and high-contrast themes.
- Consistency: Uniform styling and navigation across all pages using React component libraries.
- Responsive Design: Fully functional on desktops, tablets, and smartphones.
- User Feedback: Real-time notifications for actions like task creation, update, or failure.
- Help and Support: Integrated help tooltips, FAQ page, and contact support form.

Chapter 4

Design and Implementation

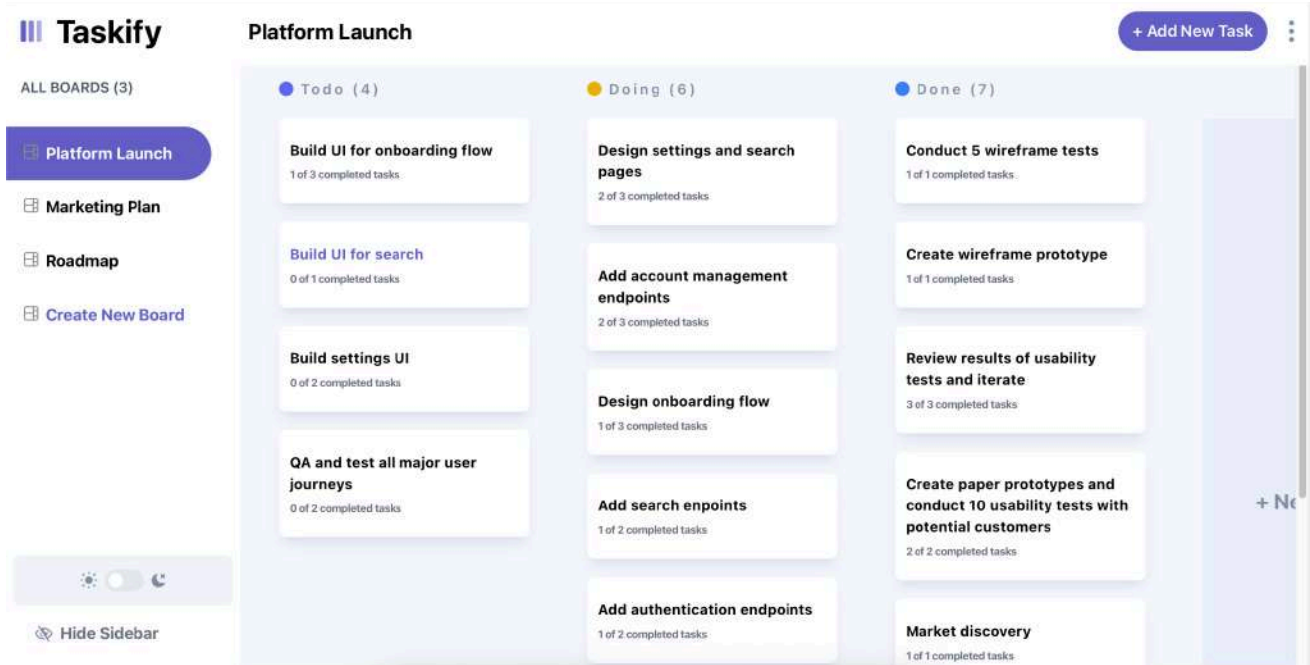


Fig:1 Website interface

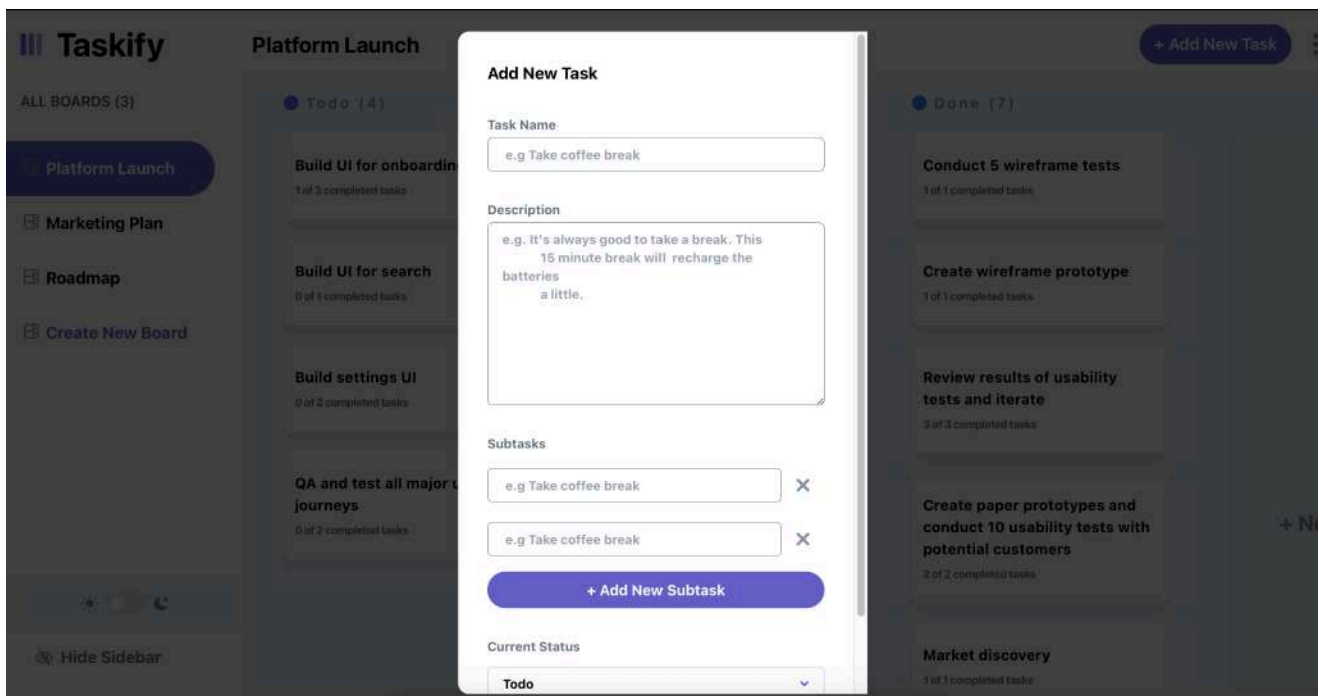


Fig:2 Adding New Task option

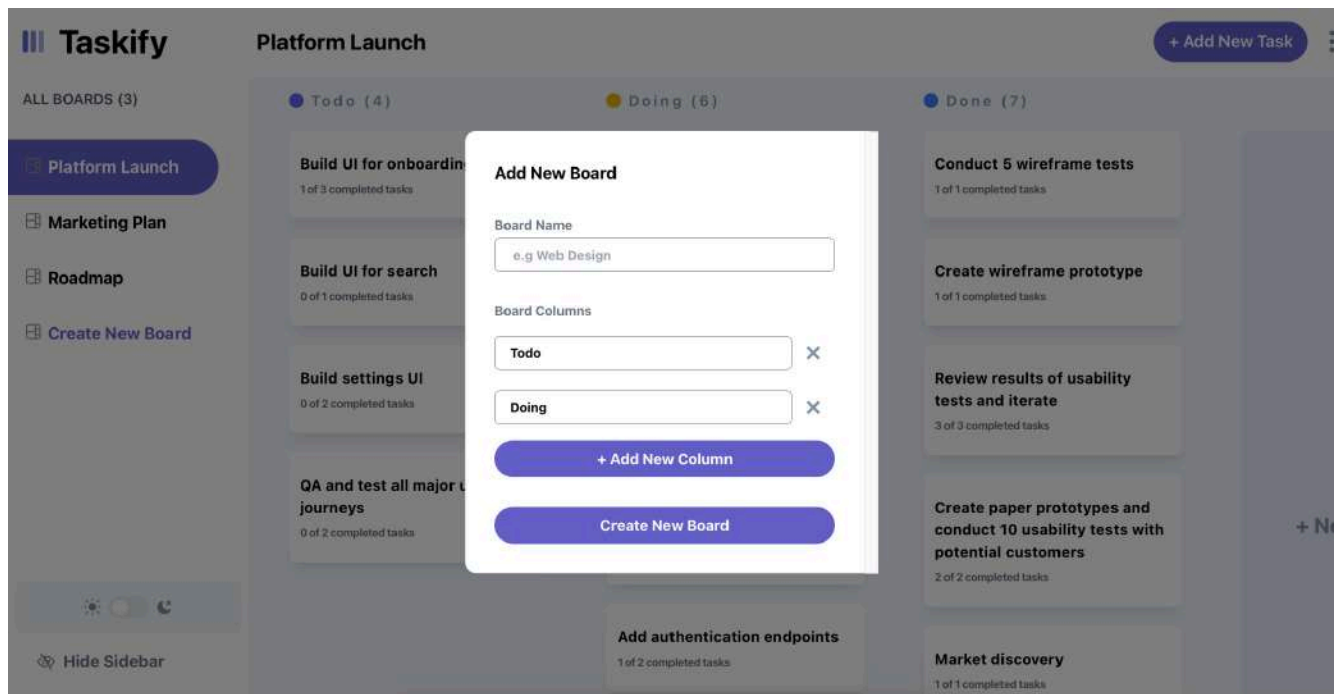


Fig:3 Adding new board option

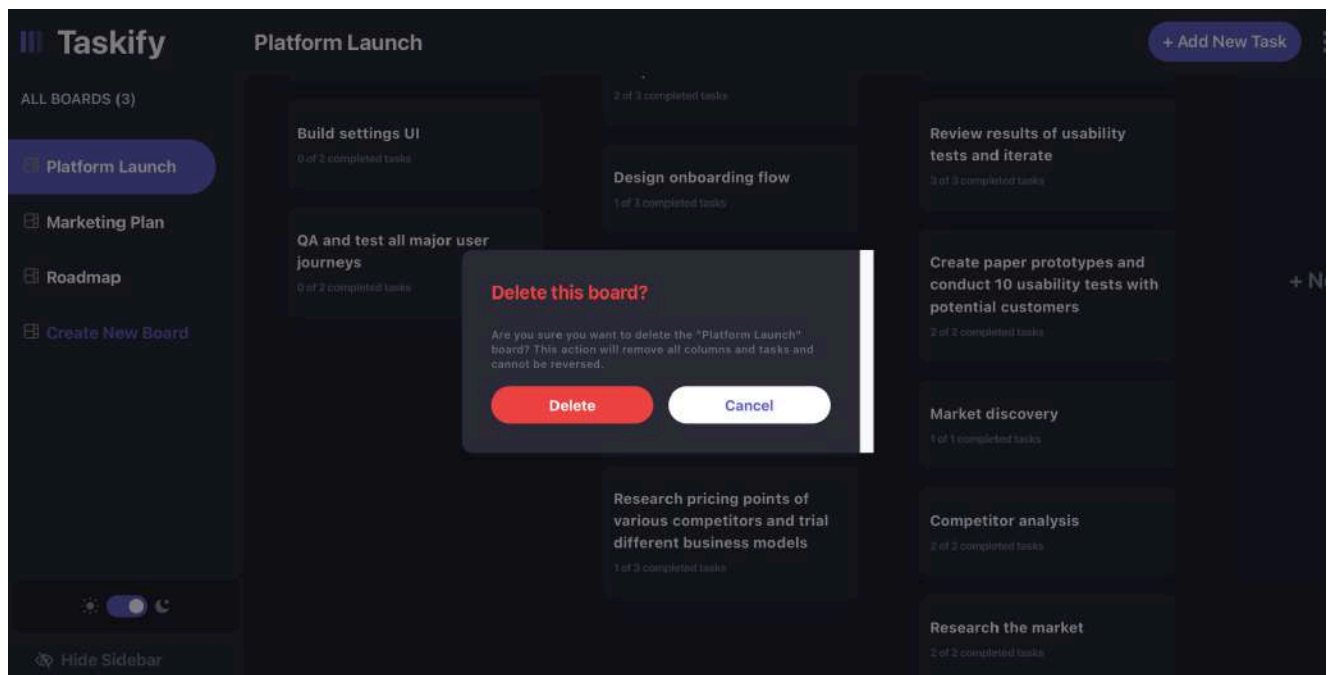


Fig:4 Delete board option

Hosting on AWS – S3

- ❖ AWS S3, or Amazon Simple Storage Service, offers reliable and scalable web hosting solutions for businesses of all sizes.
- ❖ With S3, users can easily store and retrieve static web content, such as HTML, CSS, and JavaScript files, while benefiting from high availability and low latency. Its robust security features, including encryption and access control, ensure data protection.
- ❖ Additionally, S3 integrates seamlessly with other AWS services, providing a comprehensive hosting solution for websites, applications, and content delivery networks.

The screenshot displays the 'Create bucket' interface in the AWS Management Console. The breadcrumb navigation at the top indicates the path: Amazon S3 > Buckets > Create bucket. The main heading is 'Create bucket' with an 'info' link. Below this, a note states: 'Buckets are containers for data stored in S3.'

The 'General configuration' section is expanded. It shows the 'AWS Region' as 'Asia Pacific (Mumbai) ap-south-1'. Under 'Bucket type', the 'General purpose' option is selected, with a note: 'Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.' The 'Directory' option is also visible but unselected. The 'Bucket name' field contains 'my-taskmanagement-website'. A note below the field states: 'Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). Learn More'. Below the name field, there is a section for 'Copy settings from existing bucket - optional' with a 'Choose bucket' button and a note: 'Only the bucket settings in the following configuration are copied. Format: s3://bucket/prefix'.

The 'Object Ownership' section is also visible, showing 'ACLs disabled (recommended)' selected. A note below states: 'Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.' The 'ACLs enabled' option is also visible but unselected.

The footer of the console shows 'CloudShell', 'Feedback', and copyright information: '© 2025, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences'.

Fig:5 Create S3 bucket

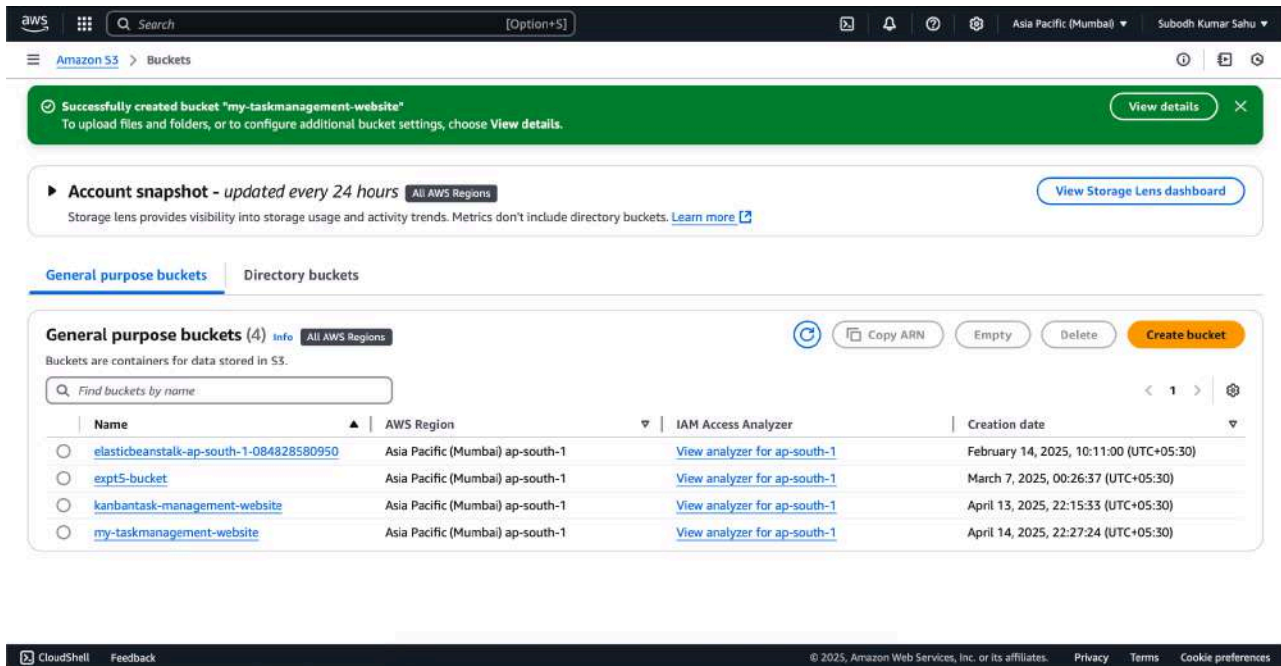


Fig:6 Bucket created successfully

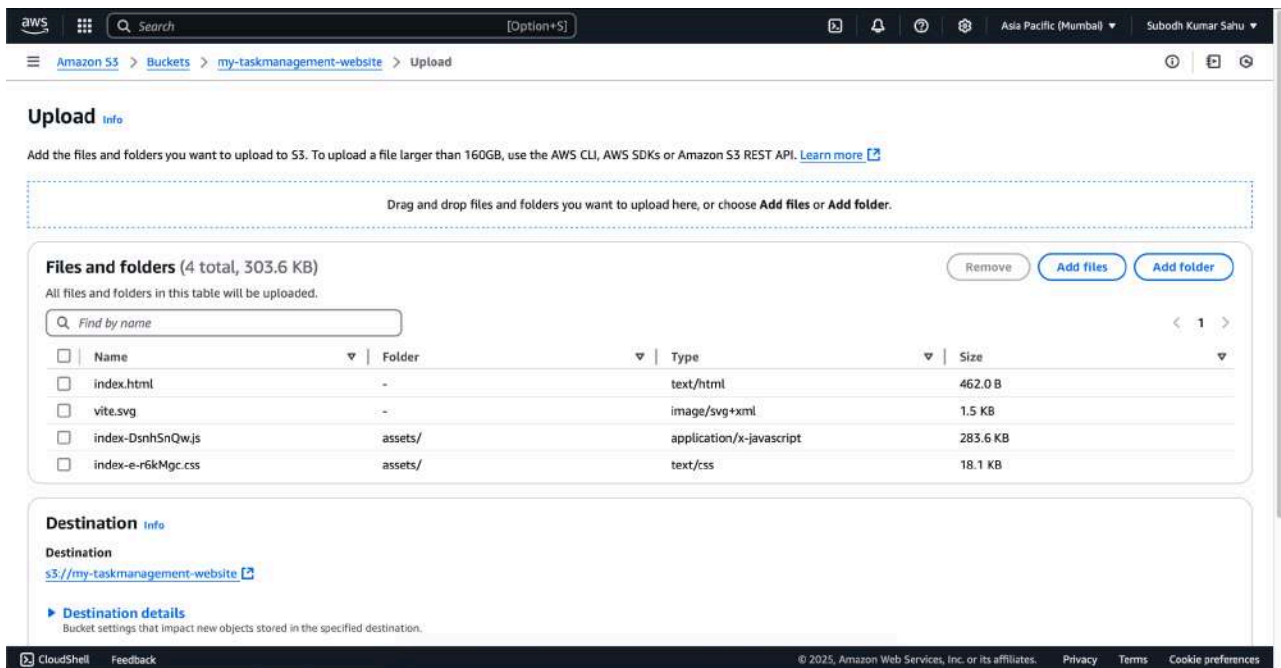


Fig:7 Upload project files to the bucket created

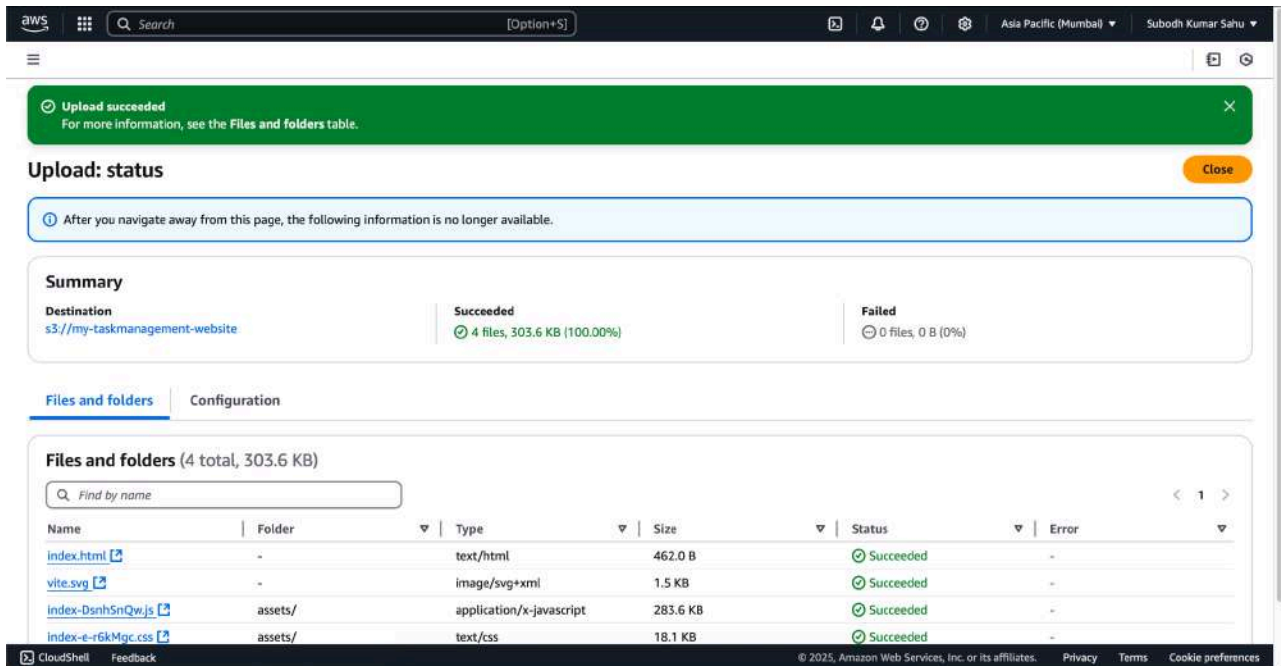


Fig:8 Projects files uploaded successfully

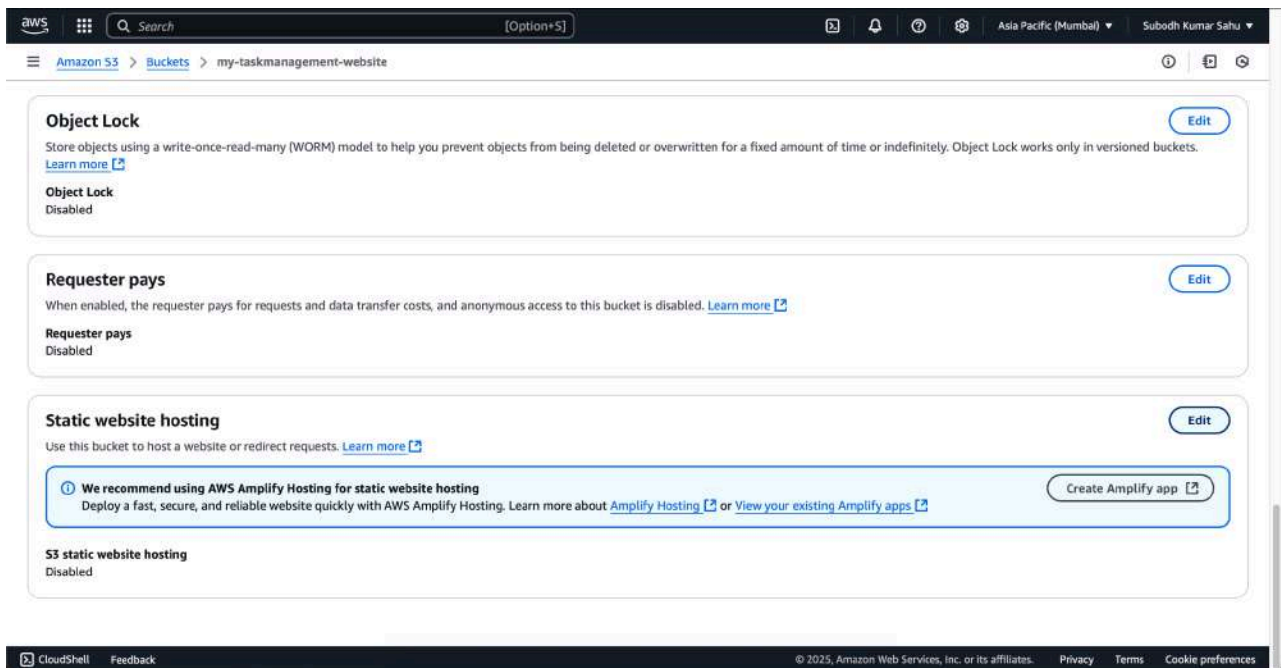


Fig:9 Make changes in the Static website hosting page

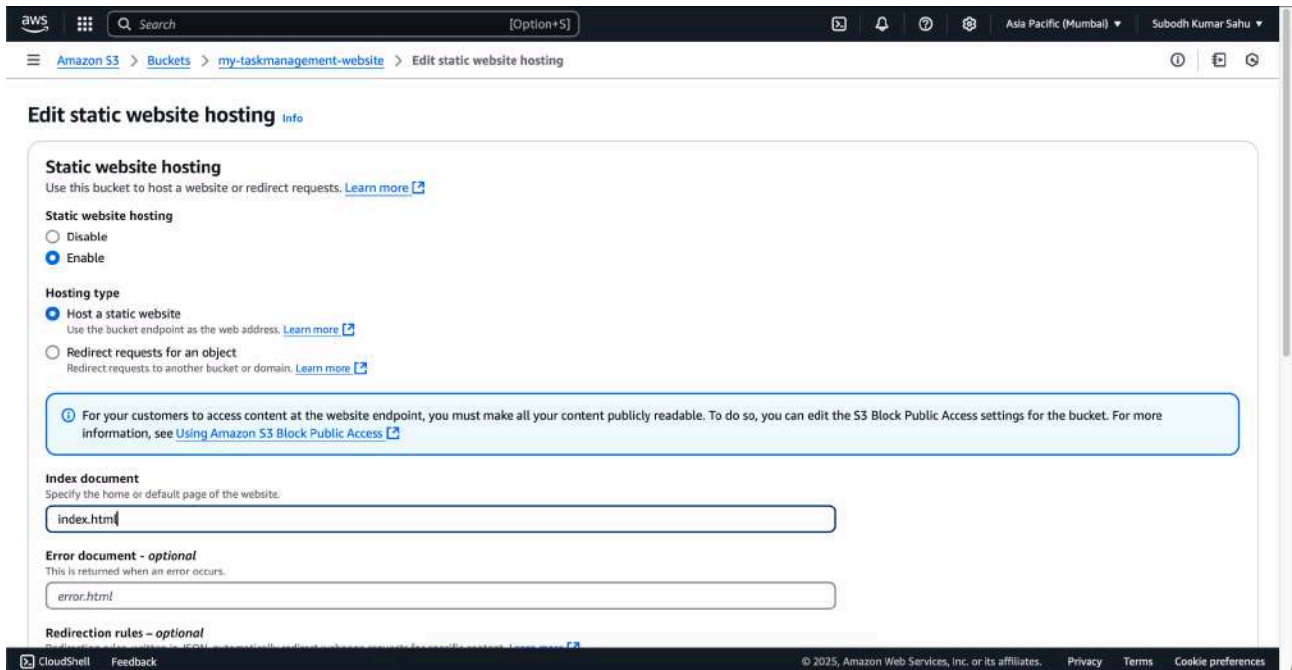


Fig:10 Enable the static website hosting

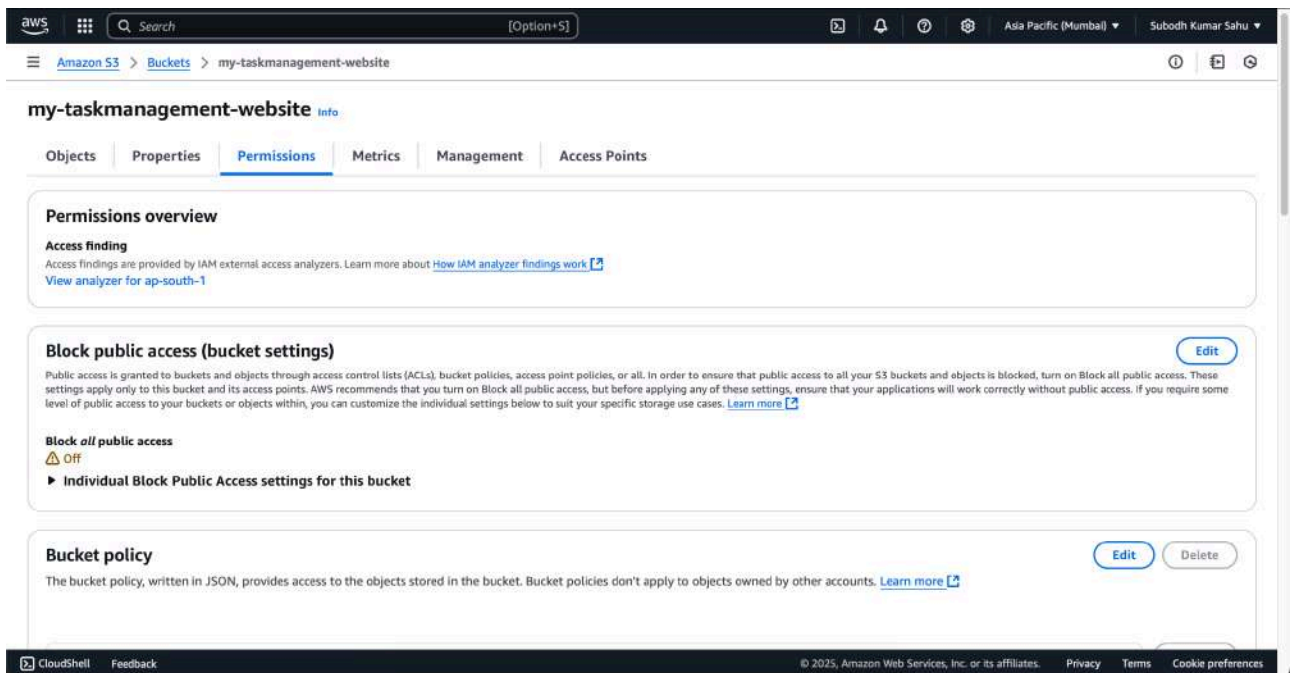
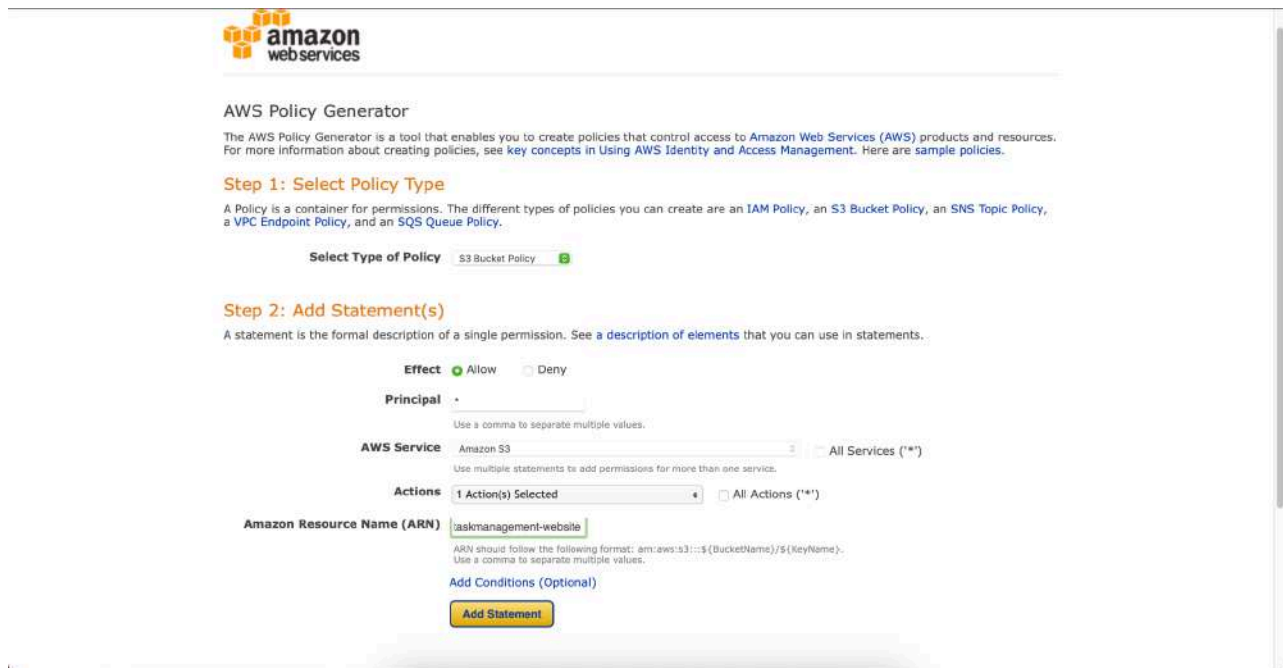


Fig:11 Edit Bucket policy and generate policy for the website



amazon web services

AWS Policy Generator

The AWS Policy Generator is a tool that enables you to create policies that control access to Amazon Web Services (AWS) products and resources. For more information about creating policies, see [key concepts in Using AWS Identity and Access Management](#). Here are [sample policies](#).

Step 1: Select Policy Type

A Policy is a container for permissions. The different types of policies you can create are an IAM Policy, an S3 Bucket Policy, an SNS Topic Policy, a VPC Endpoint Policy, and an SQS Queue Policy.

Select Type of Policy: S3 Bucket Policy

Step 2: Add Statement(s)

A statement is the formal description of a single permission. See a [description of elements](#) that you can use in statements.

Effect: ☒ Allow ☐ Deny

Principal: *

AWS Service: Amazon S3 All Services ("*")

Actions: 1 Action(s) Selected All Actions ("*")

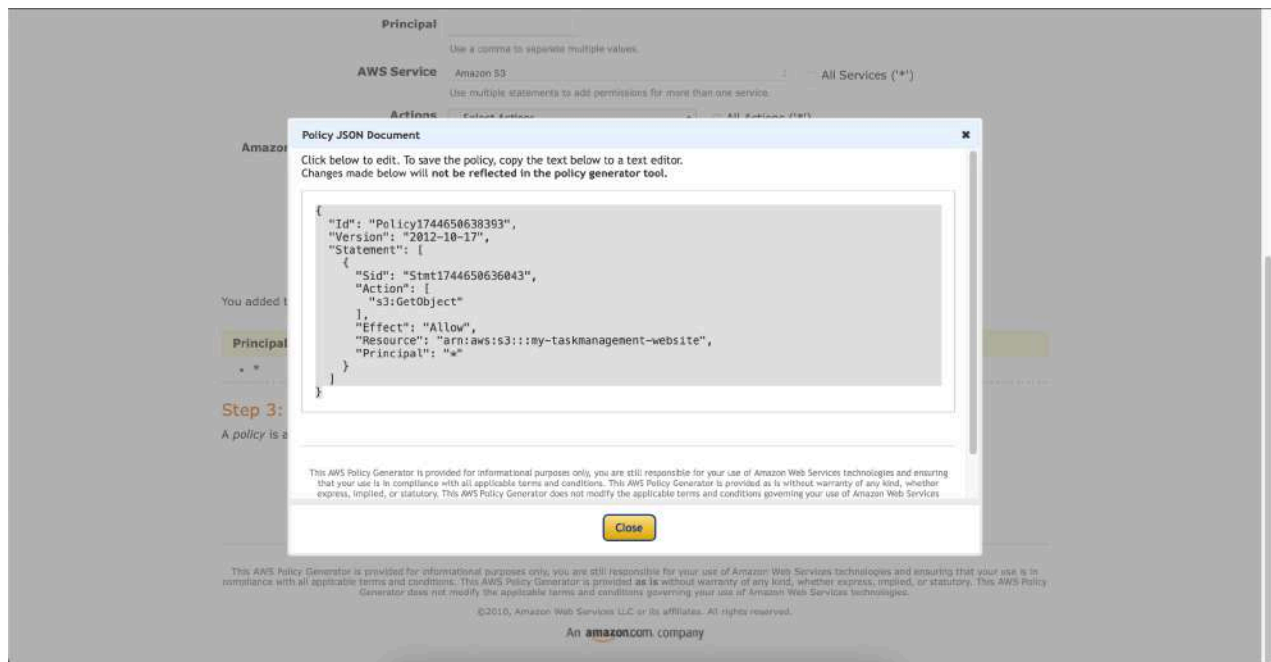
Amazon Resource Name (ARN): taskmanagement-website

ARN should follow the following format: `arn:aws:s3:::(BucketName)/$(KeyName)-`. Use a comma to separate multiple values.

[Add Conditions \(Optional\)](#)

Add Statement

Fig:12 Generate Policy for the website



Principal: *

AWS Service: Amazon S3 All Services ("*")

Actions: 1 Action(s) Selected All Actions ("*")

Amazon Resource Name (ARN): taskmanagement-website

Policy JSON Document

Click below to edit. To save the policy, copy the text below to a text editor. Changes made below will not be reflected in the policy generator tool.

```
{
  "Id": "Policy1744650638393",
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "Stmt1744650636043",
      "Action": [
        "s3:GetObject"
      ],
      "Effect": "Allow",
      "Resource": "arn:aws:s3:::my-taskmanagement-website",
      "Principal": "*"
    }
  ]
}
```

This AWS Policy Generator is provided for informational purposes only; you are still responsible for your use of Amazon Web Services technologies and ensuring that your use is in compliance with all applicable terms and conditions. This AWS Policy Generator is provided as is without warranty of any kind, whether express, implied, or statutory. This AWS Policy Generator does not modify the applicable terms and conditions governing your use of Amazon Web Services technologies.

Close

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Fig:13 Policy Generated successfully

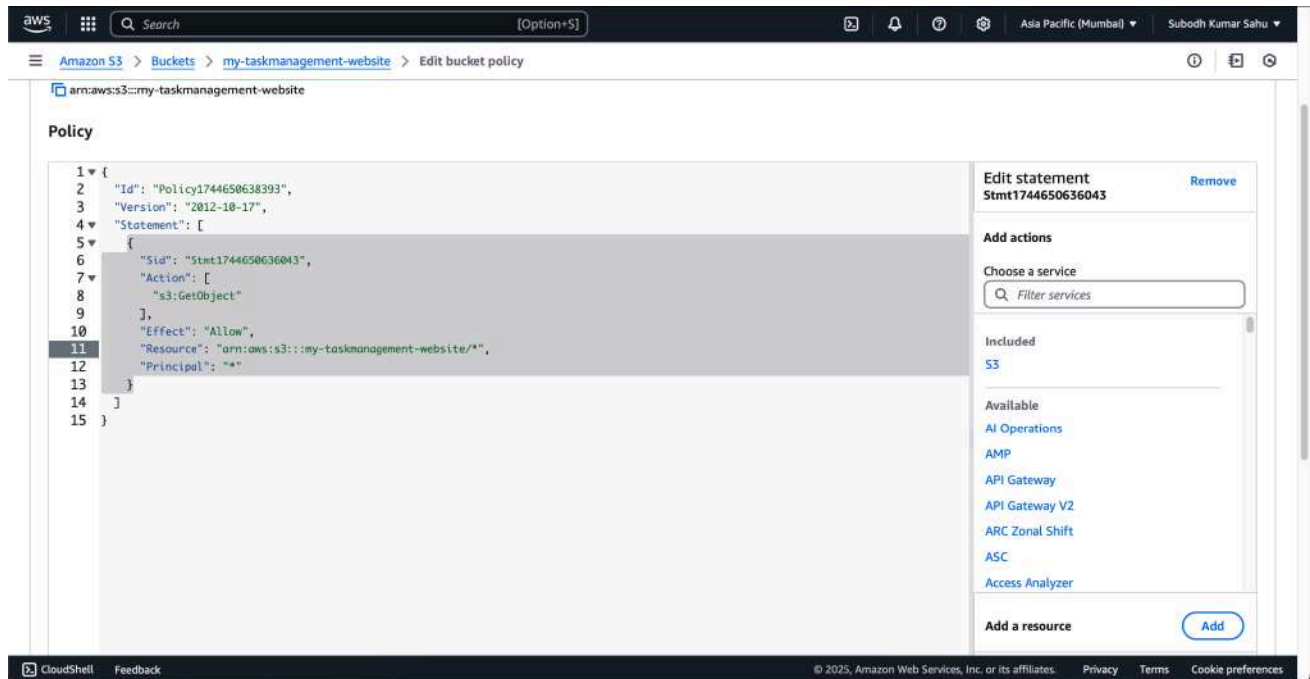


Fig:14 Add the policy that is generated and save the changes

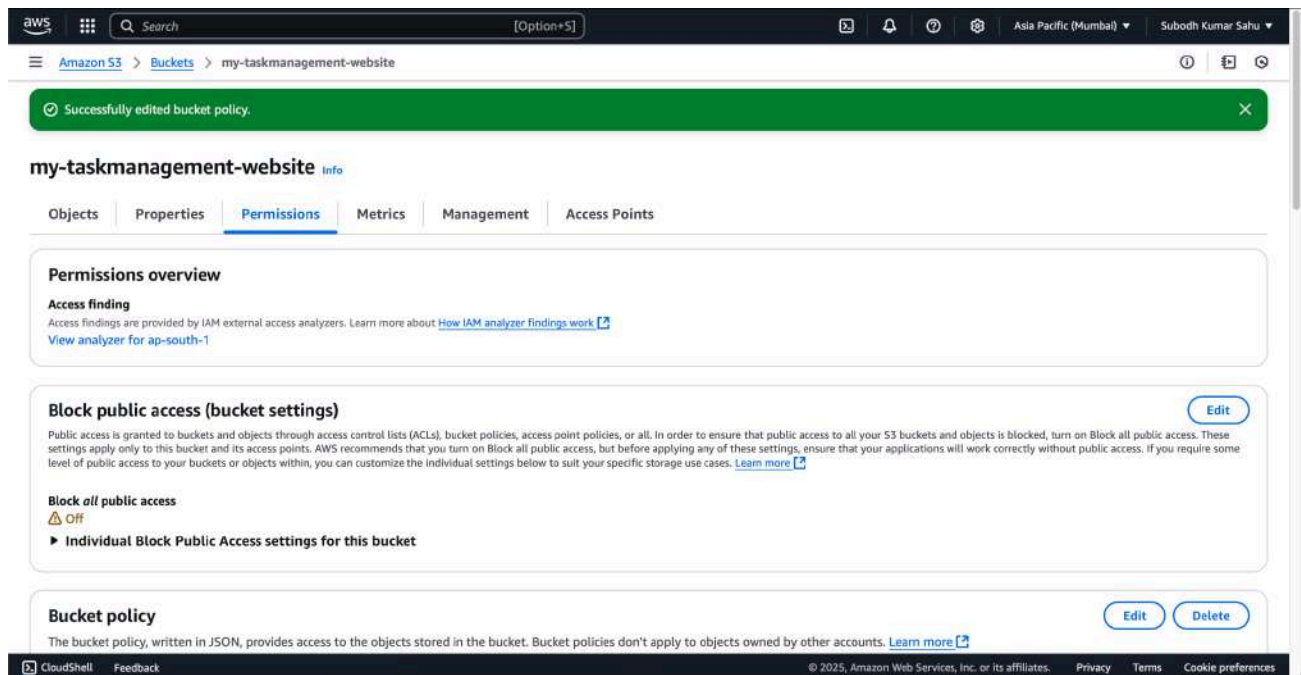


Fig:15 Policy Added successfully

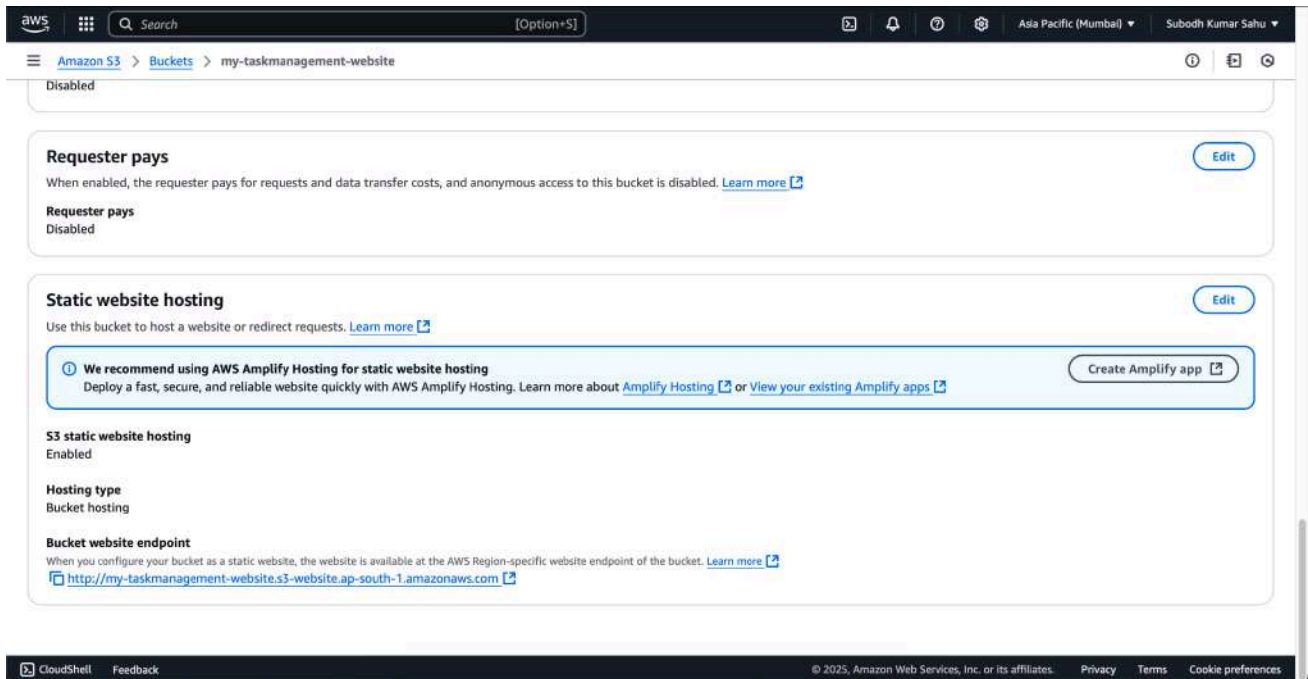


Fig:16 Finally the URL of the Website is created

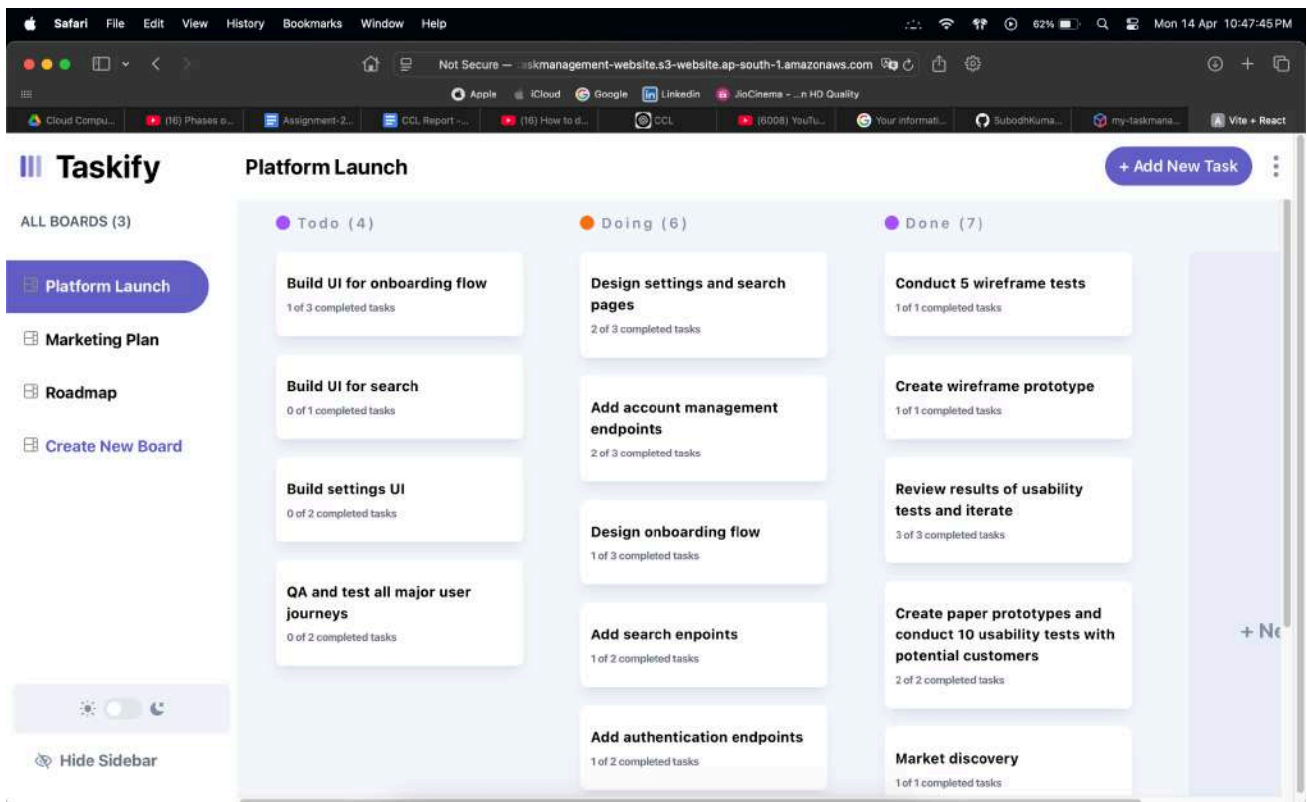


Fig:17 Finally the Website hosted on AWS server with the help of S3 bucket

Chapter 5

Conclusion and Future Scope

Conclusion

The Kanban Task Management Website provides a scalable, cloud-based solution for modern task and project management, leveraging the Kanban methodology for visual workflow tracking. By integrating powerful tools like React.js, Node.js, MongoDB, and AWS cloud services, the system ensures real-time collaboration, secure user access, and high availability. Key features such as drag-and-drop task management, role-based access, and Docker-based deployment contribute to a responsive and efficient platform. Hosting on AWS enhances the system's performance, security, and scalability—making it suitable for both small teams and large enterprises. This project successfully demonstrates the practical application of cloud computing, DevOps practices, and agile project management in a real-world scenario.

Future Scope

1. Real-Time Collaboration
 - Integrate WebSocket or Firebase for real-time updates across multiple users simultaneously.
2. Mobile App Development
 - Build a mobile version using React Native or Flutter for better accessibility.
3. Third-Party Integration
 - Integrate with tools like Slack, Google Calendar, GitHub, and Jira for enhanced productivity.
4. Analytics and Reporting
 - Add features for task analytics, performance tracking, and progress reports to help managers monitor team output.
5. Multilingual Support

- Implement internationalization (i18n) to support multiple languages and time zones.

6. AI-Based Task Recommendations

- Use machine learning to suggest task prioritization, assignment, or deadline estimation based on usage patterns.

7. Offline Mode

- Enable an offline-first approach where tasks can be edited offline and synced when the network is available.

8. Custom Workflow Templates

- Allow users to create and reuse custom workflows suited to different types of projects.

9. Advanced Security Features

- Add multi-factor authentication (MFA) and audit trails for enterprise-level security.

10. Self-Hosted or SaaS Option

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